7934 - A Deep, High Resolution NIRCam Mosaic of Cas A: Keeping Up with its Rapidly Evolving Structure

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	***Institution***
Dr. Robert A. Fesen (PI)	**Dartmouth College**
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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	NIRCam mosaic	NIRCam Imaging	(1) Cas-A

ABSTRACT

The Galactic supernova remnant Cassiopeia A (Cas A) is the youngest Galactic core-collapse SNR known and, at an estimated distance of 3.4 kpc provides us with the clearest look at the properties and explosion dynamics of a high mass supernova. Cas A's ejecta knots are typically sub-arcseconds in size and can show emission and morphological changes on timescales from a few months to a few years. No other remnant, with the exception of SN 1987A, shows such rapid optical changes across large portions of its structure.

We propose a deep NIRCam image of Cas A remnant in Cycles 4 and 5 using a filter especially sensitive to the remnant's bright emission lines to match existing WFC2 and ACS HST images but with a resolution 3-4 times better. Three other NIRCam filers will be used to match those taken in JWST's Cycle 1. These data will extend and complement the existing HST/JWST images and be an HST-JWST transition for the study and MHD

modeling of Cas A and other high mass, core-collapse SNe.

OBSERVING DESCRIPTION

This program will reproduce NIRCam mosaics obtained in Cycle 1 through program ID 1947 (PI: Milisavljevic). In addition to the three original filters F162M, F356W, and F444W, we also include the F070W that will broadly match the bandpass of existing HST/WFC F625W+F775W images. A repeat of the mosaics will be conducted in Cycle 5. These observations will be analyzed for clump sizes and compared to existing data to measure proper motion. We request that these new observations broadly match the orient of the original Cycle 1 observations to maximize overlap of the data sets.

Proposal 7934 - Targets - A Deep, High Resolution NIRCam Mosaic of Cas A: Keeping Up with its Rapidly Evolving Structure

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Cas-A	RA: 23 23 24.0000 (350.8500000d) Dec: +58 48 54.00 (58.81500d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Supernova remnants]

Observation

Proposal 7934, Observation 1: NIRCam mosaic Mon Apr 14 17:00:22 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.

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Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
3	1	10.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULL	3TIGHT	SMALL-GRID-DITHER		4

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F070W	F356W	BRIGHT1	7	1	12	12	1674.936	
2	F162M+F150W2	F444W	BRIGHT1	7	1	12	12	1674.936	

Special Requirements	
	Sequence Visits within 53.0 Days Aperture PA Range 99.92542306 to 129.92542306 Degrees (V3 100.0 to 130.0) Visits Same PA